

PhD Presentation Guide

Hybrid AE-NODE Acceleration of CFD — 15-Minute Talk

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What Your Thesis Is

You built a hybrid ML-CFD accelerator for 2D vortex shedding past a cylinder. The core idea: replace expensive WaterLily Navier-Stokes solver steps with a fast Autoencoder + Neural ODE surrogate, while using KNN-based out-of-distribution (OOD) detection to know when the ML model can no longer be trusted and the true solver must resume.

Key numbers to remember:

- Flow field: $256 \times 256 \times 2 = 131,072$ floats compressed to **16 floats** (ratio 8192:1)
- NODE network: only $\sim 1,536$ parameters — tiny and microsecond-fast
- Physics loss weights: $\lambda_{div} = 1000$, $\lambda_{curl} = 100$
- KNN threshold: 99th percentile of training distribution distances ($k = 5$)
- Hybrid switch: every 150 CFD steps, attempt an ML prediction
- Warmup / training window: $t \in [0, 17.5]$ tU/L; hybrid acceleration up to $t = 50$

Slide-by-Slide Breakdown

Slide 1 — Title

0:00–0:30

“Physics-Informed Latent Space Acceleration of Incompressible CFD
via Hybrid AE-NODE with Adaptive OOD Control”

Your name, supervisor, institution.

Slide 2 — Motivation

0:30–2:00

Hook: Turbulence simulation is one of the most computationally expensive tasks in engineering. A single $Re = 2500$ vortex shedding run on a 256×256 grid is far slower than the physical time being simulated.

- Show a single vortex shedding frame (velocity flood plot or GIF frame)
- **Problem:** Can we replace 90%+ of CFD steps with fast ML without sacrificing physical accuracy?
- **Why it’s hard:** High-dimensional fields, nonlinear chaotic dynamics, hard physical constraints (mass conservation, vorticity transport)

Slide 3 — System Overview

2:00–3:30

A single pipeline diagram (keep it simple):

CFD (WaterLily) $\rightarrow$ Snapshots $\rightarrow$ AE Encoder $\rightarrow z \in R^{16} \rightarrow$ NODE $\rightarrow \hat{z}(t + \Delta t) \rightarrow$ AE
Decoder $\rightarrow \hat{u}$

↓
KNN OOD Check → fallback to CFD if out-of-distribution

Three components: **Autoencoder** (compression), **Neural ODE** (dynamics), **KNN OOD** (safety valve). Set the architecture; next slides unpack each piece.

Slide 4 — Autoencoder: Physics-Constrained Compression 3:30–5:30

Architecture: 6-layer convolutional encoder → 16-dim latent → 6-layer bilinear-upsample decoder. Input is 4-channel: $(u, v, \mu_{0,x}, \mu_{0,y})$ — the body mask is concatenated so the network knows where the cylinder is.

Key innovation — physics losses:

Loss	Weight	Purpose
Reconstruction (L1)	$1 \times$	Field accuracy
Divergence $\ \nabla \cdot \hat{u}\ _2$	$\lambda = 1000$	Mass conservation
Curl MSE $\ \omega - \hat{\omega}\ _2$	$\lambda = 100$	Vorticity accuracy

The finite-difference stencils for divergence and curl exactly replicate WaterLily’s internal scheme — the network is penalized for violating physics in the same numerical sense as the solver it replaces.

Also mention: ghost-cell clipping (`clip_bc`) so boundary halos don’t pollute training.

Slide 5 — Neural ODE: Dynamics in Latent Space 5:30–7:30

The NODE parameterizes $dz/dt = f_\theta(z)$ with a tiny MLP (Dense $16 \rightarrow 48 \rightarrow 16$, `tanhshrink` activation), integrated with `Tsit5()`.

Multiple shooting training:

- Split time series into windows of 20 points
- Each window integrated independently from its own initial condition
- Continuity penalty (weight = 200) enforces consistency at segment boundaries
- Avoids gradient explosion over long chaotic trajectories

Show: Extrapolation plot — train region vs. test region, predicted trajectory vs. ground truth per latent dimension. Point out how the 16 latent dims capture the shedding cycle periodicity.

Slide 6 — KNN OOD Detection 7:30–9:00

Problem: A trained NODE will eventually drift outside its training distribution. Predicting there produces physically wrong results.

Solution:

1. Fit a KD-tree on training latent vectors (L2-normalised → scale-invariant metric)
2. For any new latent point: compute mean distance to $k = 5$ nearest neighbours
3. If distance $>$ 99th percentile of training distances → OOD → stop

Adaptive rollout (`predict_flex`): Don’t commit to N steps upfront. Integrate one step at a time, check OOD after each. Stop when score exceeds threshold. This is the key safety mechanism that makes the hybrid approach *reliable*, not just fast.

Slide 7 — Hybrid Runner

9:00–10:30

Four phases:

1. **Warmup** ($t = 0\text{--}17.5$ tU/L): Pure CFD, collect training snapshots
2. **Training**: Train AE $\rightarrow$ encode latents $\rightarrow$ train NODE (all in-memory)
3. **Hybrid loop** ($t = 17.5\text{--}50$ tU/L): Every 150 CFD steps, attempt ML prediction. If OOD flags ≥ 3 , extend CFD and retrain
4. **Mean-flow maintenance**: Decode intermediate latent snapshots, enforce Biot-Savart BCs, accumulate Reynolds-averaged statistics

Show: The 3-panel `plt_combined.png` — force over time with train/hybrid/retrain regions colour-coded, wall-time bars.

Slide 8 — Results

10:30–13:00

- **Speed**: Overall speedup = total reference wall time / hybrid wall time
- **Force coefficients**: Drag C_D mean error and Lift C_L RMS error vs. reference
- **Reynolds stress tensor**: $\langle u'u' \rangle$, $\langle v'v' \rangle$, $\langle u'v' \rangle$ — spatial comparison maps (`rst_comp_plot.png`)
- **Mean flow**: $\langle u \rangle$, $\langle v \rangle$ fields, hybrid vs. reference (`plt_meanflow.png`)
- **Divergence**: $\|\nabla \cdot \hat{u}\|$ near zero (physics constraint satisfied)

Key message: The hybrid system maintains turbulence statistics even during ML-accelerated periods, because mean-flow accumulation is preserved through `update_predicted_meanflow!`.

Slide 9 — Conclusions & Future Work

13:00–15:00

What you showed:

- Latent space dynamics of vortex shedding can be learned with a 16-dim NODE
- Physics-informed losses enforce mass conservation and vorticity in the compressed space
- KNN OOD detection provides a principled fallback — the system knows when it doesn't know
- Full pipeline runs online: warm up, train, accelerate, retrain if needed

Honest limitations:

- Trained on one Re; generalization requires transfer learning (TL1/TL2 already implemented)
- Fixed cylinder geometry; extension to other shapes is non-trivial
- Biot-Savart BCs must be re-imposed after each ML step — overhead cost

Future work:

- Conditional NODEs parameterised by Re (generalise across flow regimes)
- Uncertainty quantification beyond KNN (ensemble NODEs)
- 3D extension: same architecture, massively higher compression ratio

Timing Summary

Slide	Topic	Time
1	Title	0:30
2	Motivation	1:30
3	System overview	1:30
4	Autoencoder	2:00
5	Neural ODE	2:00
6	KNN OOD	1:30
7	Hybrid runner	1:30
8	Results	2:30
9	Conclusions	2:00
Total		15:00

Tips for the Committee

1. **Lead with physical intuition, not ML jargon.** Say “we compress the flow to a 16-number fingerprint and learn how those numbers evolve” before saying “autoencoder + neural ODE.”
2. **The OOD detection is your strongest differentiator.** Most ML-CFD work ignores when to trust the model. You do not. Emphasise this.
3. **Have `plt_combined.png` ready** as the single most convincing slide. Seeing the force coefficients track the reference during the hybrid phase is immediately compelling.
4. **Be ready to explain the physics loss weights** ($\lambda_{div} = 1000$, $\lambda_{curl} = 100$). Divergence violations are physically catastrophic (mass non-conservation), justifying the heavy penalty.
5. **Multiple shooting question:** Why not train the NODE end-to-end? Answer: gradient explosion over long chaotic trajectories makes direct rollout training unstable; multiple shooting is standard in scientific ML.
6. **Why 16 latent dims?** Smallest dimension that keeps reconstruction error below target (test loss ≈ 0.047). The shedding cycle has low intrinsic dimensionality.